

INDIAN INSTITUTE OF TECHNOLOGY GUWAHATI

ESTATES AND WORKS SECTION • ROADS & CIVIL INFRASTRUCTURE DIVISION
Predictive Pavement Management System — Official Work Order

Work Order ID: WO-IITG-CL007-20260926	Date Issued: 26 September 2026, 11:17 IST
Target Route: Faculty Gate Main Road	Action Type: Full Segment Resurfacing
Center GPS: 26.19533, 91.70361	Defect Cluster Count: 14 detected defects

1. PREDICTIVE ENGINEERING & RISK ASSESSMENT

This 100-meter corridor was clustered via Spatial DBSCAN combining multi-modal inputs (proactive CCTV traffic telemetry, 100 Hz bicycle accelerometer vertical shocks & crowdsourced reports). The aggregate **Road Damage Index (RDI)** is **0.417** with a cumulative surface defect area of **23.80 m²**. Compounded by pavement aging, the **Predictive Degradation Index (PDI)** **has reached 1.271**, crossing the critical threshold for sub-base deterioration. Reactive point patching is hereby discontinued for this segment in favor of predictive 100m cold-milling and asphaltic concrete resurfacing.

2. LIFECYCLE COST SAVINGS ANALYSIS (GRANICA COMPLIANT)

Metric / Parameter	Reactive Spot Patching	Predictive Full Resurfacing
Immediate Repair Cost	Rs. 105,000.00	Rs. 34,000.00
Expected Longevity	6 - 9 Months (High monsoon failure)	4 - 5 Years (Durable wear course)
Projected 3-Year Maintenance	Rs. 262,500.00	Rs. 34,000.00
Net Campus Savings	Baseline (0% Savings)	68.0% Cost Reduction

3. SEGMENT DEFECT MANIFEST

Observation ID	Source	Area (m²)	Peak Shock	PDI Score
OBS_IITG_0008	MOBILE_RUN	1.63	2.99g	1.500
OBS_IITG_0016	STUDENT_APP	1.14	1.00g	0.707
OBS_IITG_0024	CCTV	2.17	1.38g	1.493
OBS_IITG_0032	CCTV	0.65	1.55g	1.033
OBS_IITG_0040	CCTV	0.86	1.58g	0.943
OBS_IITG_0048	STUDENT_APP	2.56	1.00g	1.284

4. AUTHORIZATION & SIGN-OFF

Recommended By:

Lead Data Scientist / AI Vision Engine
IIT Guwahati Estates Automation

Approved By:

Executive Engineer (Civil)
Estates & Works Section, IIT Guwahati